



10 Available Automated Insulin Delivery (AID) Systems 2025

Available AID systems	MiniMed 780G	Tandem Control-IQ	Omnipod 5	mylife Loop	iLet Bionic Pancreas
Photo					
Firm	Medtronic (USA)	Tandem (USA)	Insulet (USA)	Cambridge AP (UK)	Beta Bionics (USA)
Indications	≥ 7 year and 8-250 U insulin per day	US: ≥ 2 years, ≥ 9-200 kg and 5-200 U insulin per day	≥ 2 year and ≥ 6 U insulin per day	≥ 1 year, >10 kg and ≥ 5 U insulin per day + pregnancy	≥ 6 year
Availability	CE-label & FDA-label	CE-label & FDA-label	CE-label & FDA-label	CE-label, FDA pending	FDA-label
CGM	Guardian 3 or 4, Simplerla Sync	Dexcom G6 or G7, FreeStyle Libre 2 Plus (future: FreeStyle Libre 3)	Dexcom G6 or G7, FreeStyle Libre 2 Plus (future: FreeStyle Libre 3)	Dexcom G6 or FreeStyle Libre 3 (Plus)	Dexcom G6 or G7, FreeStyle Libre 3 Plus
Pump	MiniMed 780G	Tandem t:slim X2, Mobi	Omnipod 5	Dana-RS, Dana-i, mylife Ypsopump	iLet ACE Pump
Mobile Control	View app on phone	t:slim X2: phone bolus, Mobi: full phone control	In US: full phone control (Android & iOS)	Full phone control (Android & iOS)	View app on phone
Target	100, 110 or 120 mg/dl (5.5, 6.1 or 6.6 mmol/l)	fixed range: 112,5-160 mg/dl (6,2-8,8 mmol/l)	110, 120, 130, 140 or 150 mg/dl (6,1-6,6-7,2-7,7-8,3 mmol/l) (adjustable per time block)	80-200 mg/dl (4,4-11,1 mmol/l) (adjustable per time block)	Usual (110 mg/dl - 6,1 mmol/l) (adjustable in 2 time blocks)
Other Targets	Temp(orary) target (target 150 mg/dl (8.3 mmol/l) without autocorrection boluses)	Sleep mode (target 112,5-120 mg/dl (6.3-6.6 mmol/l) without autocorrection boluses) Exercise Activity (target 140-160 mg/dl (7.7-8.8 mmol/l) with autocorrection boluses)	Activity Feature (target 150 mg/dl (8.3 mmol/l) and restricted insulin delivery)	Ease-off mode (target +45 mg/dl (2.6 mmol/l), less aggressive algorithm) Boost mode (insulin +35% until target, more aggressive algorithm)	Higher (130 mg/dl or 7.2 mmol/l), lower (110 mg/dl or 6.1 mmol/l)
Algorithm	Automode: PID algorithm + Fuzzy logic, based on total daily insulin dose from last 2-6 days + autocorrection boluses every 5 minutes	Control IQ: MPC algorithm that modulates basal insulin based on total daily insulin dose and weight + autocorrection boluses every hour if >180 mg/dl (>10 mmol/l)	SmartAdjust: MPC algorithm based on total daily insulin dose of the last 3 days (updates with every pod change)	Auto mode: MPC algorithm based on total daily insulin dose and weight, extended boluses every 8-12 minutes. Self-learning algorithm based on previous patterns.	iLet Dosing Decision Software: MPC algorithm based on weight. Self-learning algorithm based on previous patterns.
Adjustable Factors	Carb ratio, active insulin time (2-6h)	Basal insulin rate (different profiles possible), carb ratio, correction factor	Carb ratio (for manual boluses: correction factor, duration of insulin action, correct above threshold)	Carb ratio, Ease-off and Boost	/
Educate	Try to use target of 100 mg/dl (5.5 mmol/l) and active insulin time of 2 hours	Use sleep mode every night, use correction factor, use different basal profiles for specific activities	Wear pod and CGM in line of sight, use "Use CGM" button to take glycemic trend into account	Update body weight, avoid overusing Boost and Ease-off, avoid manual corrections	Only meal announcement needed: meal type (breakfast, lunch, or dinner) and size (usual for me, less, or more)
Unique Features	Highest TIR in (real world) studies	Many adjustable settings	First AID system with patch pump and algorithm on the pod	Algorithm on an app on your phone, low target possible, available for small children & in pregnancy	Carb counting not necessary

Available AID systems	Diabeloop	TouchCare Nano System	AndroidAPS	(Tidepool) Loop	iAPS / Trio
Photo					
Firm	Diabeloop (France)	Medtrum (China)	Open-source	Open-source	Open-source
Indications	≥ 18 year and 8-90 U insulin per day	≥ 6 year, ≥ 22 kg and ≥ 10 U insulin/day	On your own risk	On your own risk	On your own risk
Availability	CE-label, FDA pending	CE-label	/	FDA-label for earlier version (Tidepool Loop)	/
CGM	Dexcom G6	TouchCare Nano CGM	Dexcom, FreeStyle Libre, Eversense, PocTech, Glunovo, Enlite, Syai Tag, Ottai, Sibionics	Dexcom ONE/ONE+/G5/G6/G7, FreeStyle Libre 1/2, MiniMed Enlite CGM	Dexcom ONE/ONE+/G5/G6/G7, FreeStyle Libre 1/2, MiniMed Enlite CGM
Pump	Accu-Chek Insight, Kaleido, Dana-i	TouchCare Nano Pump 200 U or 300 U	Accu-Chek Combo/Insight, DanaR, Dana-i/RS, some old Medtronic pumps, Omnipod Eros/DASH, DiaconnG8, EoPatch2, Medtrum Nano/300U, Equil	Omnipod Eros; Omnipod DASH, some old Medtronic pumps	Omnipod Eros; Omnipod DASH, some old Medtronic pumps, Dana-i, Dana-RS
Mobile Control	/	Full phone control (Android & iOS)	Full phone control (only Android)	Full phone control (only iOS)	Full phone control (only iOS)
Target	100-130 mg/dl (5,5-7,2 mmol/l)	100, 110 or 120 mg/dl (5,5, 6,1 or 6,6 mmol/l)	80-200 mg/dl (4,4-11 mmol/l) (value or range, adjustable per time block)	87-180 mg/dl (4,8-10 mmol/l) (value or range, adjustable per time block)	72-180 mg/dl (4-10 mmol/l) (adjustable per time block)
Other Targets	Exercise activity & Zen mode	Exercise mode (less insulin delivery)	Customizable target and % of insulin delivery	Customizable "Overrides" of target and % of insulin delivery	Customizable target and % of insulin delivery
Algorithm	Loopmode: MPC algorithm based on total daily insulin dose, weight and average amount of carbs in a typical meal + autocorrection boluses. Self-learning algorithm based on previous patterns.	APGO: hybrid MPC/PID algorithm based on previous insulin needs and post-meal glucose, and glucose prediction. Self-learning algorithm based on previous patterns.	APS mode: MPC algorithm that modulates basal insulin based on insulin on board and carbs on board. Advanced Meal Assist or Super Micro Bolus. Extra: Autosens, Autotune, Dynamic insulin sensitivity factor	Loop: MPC algorithm that modulates basal insulin based on insulin on board and carbs on board. Temp Basal only or Automatic Bolus.	Closed Loop mode: MPC algorithm that modulates basal insulin based on insulin on board and carbs on board. Uses supermicroboluses and Autosens. Extra: Autotune, Dynamic ISF, Dynamic Carbs
Adjustable Factors	Hypo- and hyperglycemia limit; aggressivity of the algorithm at meals, normoglycemia and hyperglycemia (%), total daily insulin dose	Carb ratio	Basal insulin rate (different profiles possible), carb ratio, insulin sensitivity, duration of insulin action, insulin type, advanced algorithm features and limits	Basal insulin rate (different profiles possible), carb ratio, insulin sensitivity, insulin model, glucose safety limit, premeal range, advanced algorithm features and limits	Basal insulin rate (different profiles possible), carb ratio, insulin sensitivity, duration of insulin action, insulin model, advanced algorithm features and limits
Educate	Take rescue carbs as recommended, change total daily insuline dose if needed (higher if to much hyperglycemia; lower if to much proposed rescue carbs)	Consider additional finger stick controls because lack of safety studies	Ask someone who is using this system to help you, have a backup plan, start with time of active insuline of 5h	Ask someone who is using this system to help you, have a backup plan, use different food type icons according to absorption time (lollipop: 2h, taco 3h, pizza: 4h)	Ask someone who is using this system to help you, have a backup plan, utilize advanced settings only if you have a good understanding and have achieved stable glucose control
Unique Features	Self-learning based on previous months. Privacy mode and Zen mode. Suggests amount of rescue carbs to eat.	Optional Auto Meal Handling for meal announcements instead of carb counting	Extra RileyLink device needed if pump has no Bluetooth, you need to build the app yourself (PC needed) and complete 10 objectives (takes 2 months)	Extra RileyLink device needed if pump has no Bluetooth, you need to build the app yourself	Extra RileyLink device needed if pump has no Bluetooth, you need to build the app yourself

MiniMed 780G



Firm	Medtronic (USA)
Indications	≥ 7 year and 8-250 U insulin per day
Availability	CE-label & FDA-label
CGM	Guardian 3 or 4, Symplera Sync
Pump	MiniMed 780G
Mobile Control	View app on phone
Target	100, 110, 120 mg/dl (5,5-6,1-6,6 mmol/l)
Other Targets	Temp(orary) target (target 150 mg/dl (8.3 mmol/l) without autocorrection boluses)
Algorithm	Automode: PID algorithm + Fuzzy logic, based on total daily insulin dose from last 2-6 days + autocorrection boluses every 5 minutes
Adjustable Factors	Carb ratio, active insulin time (2-6h)
Educate	Try to use target of 100 mg/dl (5.5 mmol/l) and active insulin time of 2 hours
Unique Features	Highest TIR in real world studies

Tandem Control-IQ



Firm	Tandem (USA)
Indications	US: ≥ 2 year, 9-200 kg and 5-200 U/day Outside US: ≥ 2 year, 25-140 kg and 10-100 U/day
Availability	CE-label & FDA-label
CGM	Dexcom G6 or G7, FreeStyle Libre 2 Plus (future: FreeStyle Libre 3)
Pump	Tandem t:slim X2, Mobi
Mobile Control	t:slim X2: phone bolus, Mobi: full phone control
Target	fixed range: 112,5-160 mg/dl (6,2-8,8 mmol/l)
Other Targets	Sleep mode (target 112,5-120 mg/dl (6.3-6.6 mmol/l) without autocorrection boluses) Exercise Activity (target 140-160 mg/dl (7.7-8.8 mmol/l) with autocorrection boluses)
Algorithm	Control IQ: MPC algorithm that modulates basal insulin based on total daily insulin dose and weight + autocorrection boluses every hour if >180 mg/dl (>10 mmol/l)
Adjustable Factors	Basal insulin rate (different profiles possible), carb ratio, correction factor
Educate	Use sleep mode every night, use correction factor, use different basal profiles for specific activities
Unique Features	Many adjustable settings

Omnipod 5



Firm	Insulet (USA)
Indications	≥ 2 year
Availability	CE-label & FDA-label
CGM	Dexcom G6 or G7, FreeStyle Libre 2 Plus (future: FreeStyle Libre 3)
Pump	Omnipod 5
Mobile Control	Full phone control (Android & iOS)
Target	110, 120, 130, 140 or 150 mg/dl (6,1-6,6-7,2-7,7-8,3 mmol/l) (adjustable per time block)
Other Targets	Activity Feature (target 150 mg/dl (8.3 mmol/l) and restricted insulin delivery)
Algorithm	SmartAdjust: MPC algorithm based on total daily insulin dose of the last 3 days (updates with every pod change)
Adjustable Factors	Carb ratio (for manual boluses: correction factor, duration of insulin action, correct above threshold)
Educate	Wear pod and CGM in line of sight, use "Use CGM" button to take glycemic trend into account
Unique Features	First AID system with patch pump and algorithm on the pod



Firm	Cambridge AP (UK)
Indications	≥ 1 year and > 10 kg and ≥ 5 U insulin per day + pregnancy
Availability	CE-label, FDA pending
CGM	Dexcom G6 or FreeStyle Libre 3 (Plus)
Pump	mylife YpsoPump (Dana-RS, Dana-i)
Mobile Control	Full phone control (Android & iOS)
Target	80-200 mg/dl (4,4-11,1 mmol/l) (adjustable per time block)
Other Targets	Ease-off mode (target +45 mg/dl (2.6 mmol/l), less aggressive algorithm) Boost mode (insulin +35% until target, more aggressive algorithm)
Algorithm	Auto mode: MPC algorithm based on total daily insulin dose and weight, extended boluses every 8-12 minutes. Self-learning algorithm based on previous patterns.
Adjustable Factors	Carb ratio, Ease-off and Boost
Educate	Update body weight, avoid overusing Boost and Ease-off, avoid manual corrections
Unique Features	Algorithm on an app on your phone, low target possible, available for small children & in pregnancy

iLet Bionic Pancreas



Firm	Beta Bionics (USA)
Indications	≥ 6 year
Availability	FDA-label
CGM	Dexcom G6 or Dexcom G7, FreeStyle Libre 3 plus
Pump	iLet ACE Pump
Mobile Control	View app on phone
Target	Usual (110 mg/dl - 6,1 mmol/l) (adjustable in 2 time blocks)
Other Targets	Higher (130 mg/dl or 7.2 mmol/l), lower (110 mg/dl or 6.1 mmol/l)
Algorithm	iLet Dosing Decision Software: MPC algorithm based on weight. Self-learning algorithm based on previous patterns.
Adjustable Factors	/
Educate	Only meal announcement needed: meal type (breakfast, lunch, or dinner) and size (usual for me, less, or more)
Unique Features	Carb counting not necessary

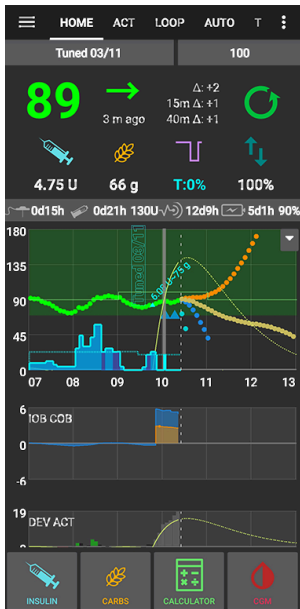


Firm	Diabeloop (France)
Indications	≥ 18 year and 8-90 U insulin per day
Availability	CE-label, FDA pending
CGM	Dexcom G6
Pump	Accu-Chek Insight, Kaleido
Mobile Control	/
Target	100-130 mg/dl (5,5-7,2 mmol/l)
Other Targets	Exercise activity (target +70 mg/dl (3.8 mmol/l), less aggressive algorithm) Zen mode (target +10-40 mg/dl (+0.5-2.2 mmol/l), less aggressive algorithm)
Algorithm	Loopmode: MPC algorithm based on total daily insulin dose, weight and average amount of carbs in a typical meal + autocorrection boluses. Self-learning algorithm based on previous patterns.
Adjustable Factors	Hypo- and hyperglycemia limit; aggressivity of the algorithm at meals, normoglycemia and hyperglycemia (%), total daily insulin dose
Educate	Take rescue carbs as recommended, change total daily insuline dose if needed (higher if to much hyperglycemia; lower if to much proposed rescue carbs)
Unique Features	Self-learning based on previous months. Privacy mode and Zen mode. Suggests amount of rescue carbs to eat.

TouchCare Nano System

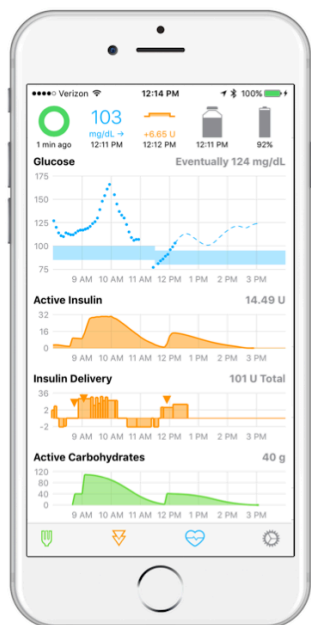


Firm	Medtrum (China)
Indications	≥ 6 year, ≥ 22 kg and ≥ 10 U insulin/day
Availability	CE-label
CGM	TouchCare Nano CGM
Pump	TouchCare Nano Pump 200 U or 300 U
Mobile Control	Full phone control (Android & iOS)
Target	100, 110 or 120 mg/dl (5.5, 6.1 or 6,6 mmol/l)
Other Targets	Exercise mode (less insulin delivery)
Algorithm	APGO: hybrid MPC/PID algorithm based on previous insulin needs and post-meal glucose, and glucose prediction. Self-learning algorithm based on previous patterns.
Adjustable Factors	Carb ratio
Educate	Consider additional finger stick controls because lack of safety studies
Unique Features	Optional Auto Meal Handling for meal announcements instead of carb counting

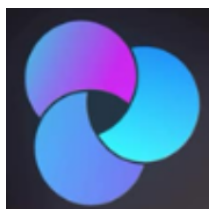
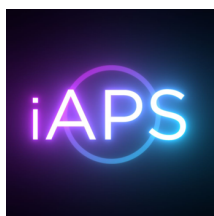
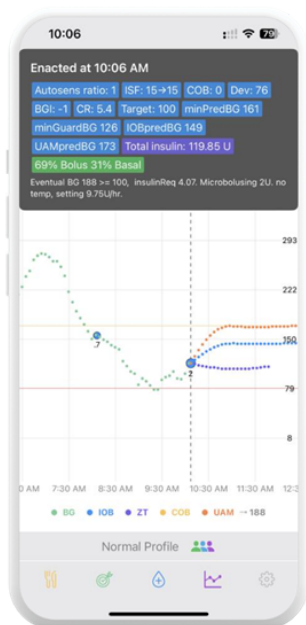


Firm	Open-source
Indications	On your own risk
Availability	/
CGM	Dexcom, FreeStyle Libre, Eversense, PocTech, Glunovo, Enlite, Syai Tag, Ottai, Sibionics
Pump	Accu-Chek Combo/Insight, DanaR, Dana-i/RS, some old Medtronic pumps, Omnipod Eros/DASH, DiaconnG8, EoPatch2, Medtrum Nano/300U, Equil
Mobile Control	Full phone control (only Android)
Target	80-200 mg/dl (4,4-11 mmol/l) (value or range, adjustable per time block)
Other Targets	Customizable target and % of insulin delivery
Algorithm	APS mode: MPC algorithm that modulates basal insulin based on insulin on board and carbs on board. Advanced Meal Assist or Super Micro Bolus. Extra: Autosens, Autotune, Dynamic insulin sensitivity factor
Adjustable Factors	Basal insulin rate (different profiles possible), carb ratio, insulin sensitivity, duration of insulin action, insulin type, advanced algorithm features and limits
Educate	Ask someone who is using this system to help you, have a backup plan, start with time of active insuline of 5h
Unique Features	Extra RileyLink device needed if pump has no Bluetooth, you need to build the app yourself (PC needed) and complete IO objectives (takes 2 months)

(Tidepool) Loop



Firm	Open-source
Indications	On your own risk
Availability	FDA-label for earlier version (Tidepool Loop)
CGM	Dexcom ONE/ONE+/G5/G6/G7, FreeStyle Libre 1/2, MiniMed Enlite CGM
Pump	Omnipod Eros; Omnipod DASH, some old Medtronic pumps
Mobile Control	Full phone control (only iOS)
Target	87-180 mg/dl (4,8-10 mmol/l) (value or range, adjustable per time block)
Other Targets	Customizable "Overrides" of target and % of insulin delivery
Algorithm	Loop: MPC algorithm that modulates basal insulin based on insulin on board and carbs on board. Temp Basal only or Automatic Bolus.
Adjustable Factors	Basal insulin rate, carb ratio, insulin sensitivity, insulin model, glucose safety limit, premeal range, advanced algorithm features and limits
Educate	Ask someone who is using this system to help you, have a backup plan, use different food type icons according to absorption time (lollipop: 2h, taco 3h, pizza: 4h)
Unique Features	Extra RileyLink device needed if pump has no Bluetooth, you need to build the app yourself



Firm	Open-source
Indications	On your own risk
Availability	/
CGM	Dexcom ONE/ONE+/G5/G6/G7, FreeStyle Libre 1/2, MiniMed Enlite CGM
Pump	Omnipod Eros; Omnipod DASH, some old Medtronic pumps, Dana-i, Dana-RS
Mobile Control	Full phone control (only iOS)
Target	72-180 mg/dl (4-10 mmol/l) (adjustable per time block)
Other Targets	Customizable target and % of insulin delivery
Algorithm	Closed Loop mode: MPC algorithm that modulates basal insulin based on insulin on board and carbs on board. Uses supermicroboluses and Autosens. Extra: Autotune, Dynamic ISF, Dynamic Carbs
Adjustable Factors	Basal insulin rate (different profiles possible), carb ratio, insulin sensitivity, duration of insulin action, insulin model, advanced algorithm features and limits
Educate	Ask someone who is using this system to help you, have a backup plan, utilize advanced settings only if you have a good understanding and have achieved stable glucose control
Unique Features	Extra RileyLink device needed if pump has no Bluetooth, you need to build the app yourself



Thank You!

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